

Introduction for students: why Part 2 of Chapter 4 matters

Part 1 gave you methods that solve any circuit. Part 2 gives you tools that make a circuit simpler before you solve it, and that answer the questions engineers actually ask: what does a load see, what happens when one source changes, and how much power can a source deliver? Part 2 runs from week 7 to week 9, and the Part 2 quiz is at the start of week 10.

What you will learn in Part 2

- **Source transformations:** a voltage source in series with a resistor can be replaced by a current source in parallel with the same resistor, and back again. A chain of these reduces a circuit step by step.
- **Superposition:** in a linear circuit with several independent sources, any current or voltage is the sum of the contributions of each source acting alone. Powers do not add this way.
- **Thévenin equivalent:** seen from two terminals, any linear circuit behaves like one voltage source V_{Th} in series with one resistor R_{Th} . V_{Th} is the open-circuit voltage; $R_{Th} = V_{Th}/i_{sc}$ (use a test source when both are zero).
- **Norton equivalent:** the same circuit seen as a current source $I_N = i_{sc}$ in parallel with R_{Th} .
- **Finding R_{Th} :** deactivate the independent sources and combine resistors, or, when dependent sources are present, apply a test source.
- **Maximum power transfer:** a load receives the most power when its resistance equals R_{Th} , and that power is $(V_{Th})^2/(4R_{Th})$.

Later in this course

Topic	Where Part 2 comes back
Chapter 5: Operational amplifiers	The source driving an amplifier and the load it drives are both modelled as Thévenin equivalents
Chapter 7: First-order circuits	A capacitor or inductor sees the rest of the circuit as one Thévenin equivalent; the time constant $\tau = R_{Th} C$ or L/R_{Th} uses the Thévenin resistance seen by the capacitor or inductor, often found with a test source

In your program

- **Electrical and Electronics:** a power system engineer models the whole grid behind a substation as a Thévenin equivalent to work out what a fault or a new load will do; an amplifier's input and output are specified by their Thévenin resistances.
- **Computer, Communication and Telecom:** antennas, cables and receivers are matched so that maximum power reaches the receiver, the same condition $R_L = R_{Th}$ in its general form.
- **Biomedical:** an electrode on the skin is a Thévenin source with a large R_{Th} , which is why the amplifier reading it needs an even larger input resistance.
- **Mechanical and Industrial:** a sensor and its wiring form a Thévenin source; the reading drops when the instrument draws current.
- **Surveying:** the battery pack of a field instrument is an open-circuit voltage behind an internal resistance, a Thévenin equivalent; that is why its voltage sags when the instrument draws current.

As an engineer

- **Reduce before you solve.** One Thévenin equivalent replaces a whole network when only one load changes, so each new load takes one line instead of a new set of equations.
- **Measure two points, know the source.** The open-circuit voltage and the voltage under one known load are enough to find V_{Th} and R_{Th} of a real source you cannot open.
- **Maximum power is not maximum efficiency.** At maximum power transfer the load receives only half the power of the Thévenin source, and in the real circuit often less (S10: 32%). Power systems run far from this point; signal circuits often sit on it.

In real life

- **Your generator subscription.** The socket voltage falls each time you switch on a heater. Two readings, with nothing on and with one heater on, give the Thévenin equivalent of the generator and its cables, and predict the voltage with any load (S15).
- **A power bank or phone charger.** Its open-circuit voltage and internal resistance decide how fast your phone charges, and why a long, thin cable can slow it down.

How to study this part, and where AI fits

- AI tools are allowed on the practice sheet. Use them to check your work, not to replace it.
- In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many (arXiv 2512.10159)¹. In this part these slips show up as a current-source arrow pointing the wrong way after a source transformation, a Thévenin source drawn upside down, or a current from one source added with the wrong sign in superposition. S11 shows such a slip, and S12 another common one, a current source deactivated as a short circuit: find them before you trust any answer.
- Check every answer yourself: V_{Th}/R_{Th} must equal the short-circuit current, R_{Th} found by deactivation must agree with V_{Th}/i_{sc} , and the powers must add to zero.
- The quiz is device-free, and every question is a twin of a practice-sheet question. If you can do the sheet on your own, you will do well on the quiz.

Checklist: you can do these without help

- ☐ Transform a voltage source with a series resistor into a current source with a parallel resistor, with the arrow the right way, and back
- ☐ Use superposition to find a current, and explain why powers cannot be added
- ☐ Find V_{Th} , i_{sc} and R_{Th} , and draw the Thévenin and Norton equivalents
- ☐ Find R_{Th} by deactivating the independent sources, and with a test source when the circuit has a dependent source
- ☐ Choose the load for maximum power transfer and find that power and the efficiency
- ☐ Check an equivalent with $i_{sc} = V_{Th}/R_{Th}$

¹<https://arxiv.org/html/2512.10159>